

1. What is the derivative of x squared?

- a) x
- b) $2x$
- c) 2
- d) x squared

Answer: b

2. The limit of $(\sin x) / x$ as x approaches 0 is:

- a) 0
- b) 1
- c) infinity
- d) undefined

Answer: b

3. The derivative of a constant is:

- a) 1
- b) 0
- c) the constant itself
- d) infinity

Answer: b

4. The integral of x with respect to x is:

- a) x squared / 2 + C
- b) x squared + C
- c) $2x$ + C
- d) $1 / x$ + C

Answer: a

5. The chain rule is used to differentiate:

- a) composite functions
- b) polynomials only
- c) constants
- d) matrices

Answer: a

6. The derivative of e to the power x is:

- a) x times e to the power x
- b) e to the power x
- c) 1
- d) $\ln x$

Answer: b

7. A differentiable function has a local maximum where its derivative:

- a) is greater than zero
- b) equals zero
- c) is less than zero
- d) is undefined only

Answer: b

8. The integral of $1/x$ with respect to x is:

- a) $1/x$ squared
- b) $\ln |x| + C$
- c) x
- d) e to the power x

Answer: b

9. The derivative of $\sin x$ is:

- a) $\cos x$
- b) $-\cos x$
- c) $\sin x$
- d) $\tan x$

Answer: a

10. A definite integral gives the:

- a) area under a curve
- b) slope of a curve
- c) tangent line
- d) radius of curvature

Answer: a

11. State the fundamental theorem of calculus.

(Write your answer in the space provided.)

12. Differentiate $y = x^3 + 2x^2 - 5x + 7$.

(Write your answer in the space provided.)